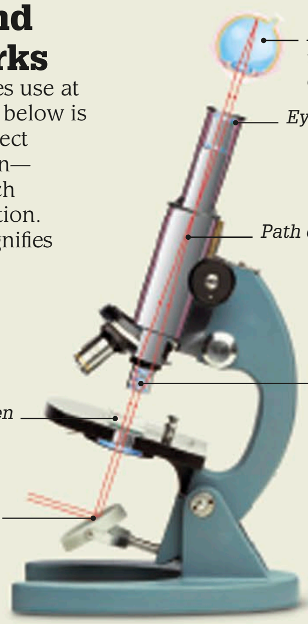
Leeuwenhoek's drawings of microscopic life

Single-lens microscope

Dutch scientist Antoni van Leeuwenhoek was able to achieve greater magnifications with his single-lens microscopes than Robert Hooke did with his compound microscope. Leeuwenhoek ground all his lenses himself, some of them no bigger than a pinhead.

How a compound microscope works

Compound light microscopes use at least two lenses. Light from below is reflected up through the object being viewed—the specimen—into the objective lens, which produces the first magnification. The eyepiece lens then magnifies the image again.

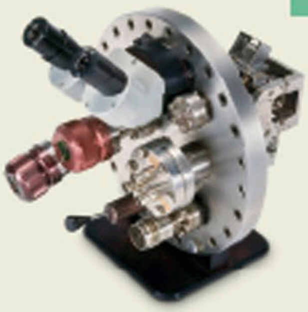


The world magnified

Today, there are three kinds of microscope. Researchers use light, or optical, microscopes to view biological specimens such as cells and tissue. Electron microscopes, including the scanning tunneling microscope, which use a beam of electrons to reveal an image, can look at much smaller things in very great detail.

2,000 times
This 19th-century light microscope can magnify objects up to 2,000 times. Its achromatic lenses create a sharper image by focusing different color wavelengths together.

10 million times
Electron microscopes fire a beam of electrons at a specimen contained in a vacuum. This example, dating from around 1946, was one of the first to be mass-produced. Modern versions can magnify up to 10 million times.



1 billion times
The scanning tunneling microscope (STM) uses a sharp metal probe to scan the surface of an object at an atomic level, allowing scientists to "see" individual atoms. The atomic force microscope works in a similar way.

1880s

Working for German instrument-maker Carl Zeiss, German optical scientist Ernst Abbe made radical improvements to microscope design.

1882

German microbiologist Robert Koch developed ways of staining bacteria with violet dye to make them more visible under a microscope.

1903

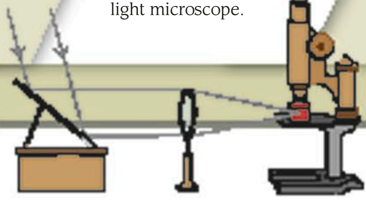
German chemist Richard Zsigmondy built the ultramicroscope, which let him view objects that could not be seen with a light microscope.

1931

German physicist Ernst Ruska invented the first scanning electron microscope (SEM) that used electron beams to create images.

1981

The scanning tunneling microscope was the first that allowed scientists to see at an extremely small scale, down to a nanometer (a billionth of a meter).



Zsigmondy's ultramicroscope